CST 3513 – Java OOP Study Plan

2-day crash plan · about 4 hours per day · built around your course folder

The 4 methods behind the plan

1. **Past exams first.** Take them cold, on paper, timed. Misses tell you what to study.
2. **Write code by hand from memory.** Paper exams test recall of syntax, not recognition.
3. **Trace "what does this print?"** Predict output on paper, then run it to check.
4. **One-page cheat sheet.** Every mistake becomes one rule in your own words. Review it repeatedly.

Day 1 – Learn the core, in order

Block 1: Diagnose (45 min)

1. Open **6 Exams & Quizzes**. Take `CST3513_F26_M.doc` on paper: no notes, 30-minute limit, leave blanks where stuck.
2. Mark it with `CST3513_F26_M_S.doc` and `M_Solution.txt`.
3. List the question types you missed (constructors, what prints, inheritance...).

Block 2: Classes, constructors, static (75 min)

1. Skim **7 Simplified Slides**: `4_Classes_and_Objects` and `5_Static_and_References` (10 min each).
2. In **4 Code Examples**, for `Clock_Class`, `Time_class`, `Class_Private_Pm`, `Static_class_D`:
   - Read for 2 minutes, cover it, rewrite by hand, compare, circle mistakes.
3. Start your cheat sheet:
   - Class template: private fields, constructor, getters/setters, `toString`
   - Constructor overloading and `this(...)`
   - Static vs instance (a static method can't use instance fields)
   - Reference vs primitive behavior

Block 3: Inheritance (75 min)

1. Read `7 Simplified Slides/6_Inheritance`.
2. Rewrite from memory: `Inheritance_Vehicle_Truck`, `Inheritance_TwoDShape_class`, `Rectangle_Cube_Inheritance1`.
3. Add to your cheat sheet:
   - `extends`; `super(...)` must be the first line of a child constructor; `super.method()` calls the parent version
   - Parent constructor runs before the child's; in a multilevel hierarchy, constructors run top-down
   - Private parent fields are reached only via getters/setters; `protected` is visible in subclasses

Block 4: Overriding and polymorphism (60 min)

1. Read `Polymorphism_def.docx` and `Overridden Methods_class.docx` (2 Lecture Notes).
2. Tracing practice with `Overridden_Methods_1/2/3` and `Polymorphism_Rectangle`: write the printed output by hand first, then run and compare. Each miss becomes a one-line rule.
3. Add to your cheat sheet:

  - Overloading = same name, different parameters, decided at compile time
  - Overriding = same signature in a child class, decided at run time by the object's actual type
  - `Parent p = new Child();` runs the child's overridden methods, but only parent-declared methods are callable on p

Day 2 – Exam simulation and weak spots

Block 5: Second exam practice (75 min)

1. Take `CST3513_F26_T1.doc` on paper, timed, cheat sheet closed.
2. Mark against `CST3513_F26_T1_S.doc`.
3. Spend 20 minutes re-studying only what you got wrong.

Block 6: Code-writing drills (60 min)

1. Write on paper with no files open: a `Student` or `Account` class (constructor, getters/setters, `toString`, one real method).
2. A `Shape` parent with `Rectangle` and `Circle` children, each overriding `area()`.
3. A `main` that stores objects in a parent-type array and loops printing `toString()`.
4. Compare each with your folder's code and fix anything missing.

Block 7: Review papers, arrays and methods (60 min)

- Do the `R_CST3513_Midterm_F26` and `R_CST3513_F26_T1` papers, only on topics where you're still shaky.
- If they were on your weak list, spend 20 min on `2_Methods` and `3_Arrays` (arrays as arguments, return values, loops over arrays).

Block 8: Final review (30 min, night before)

- Read your cheat sheet twice.
- Rewrite two programs from memory.
- Skim the `Overridden_Methods` and `Polymorphism` outputs once more – easy points to lose.

Rules to follow throughout

- **Always write by hand.** The exam is on paper.
- **No passive rereading.** Recall, trace or write instead.
- **One mistake = one rule** on your cheat sheet; don't redo whole topics.
- **Sleep before the exam.** It matters more than one extra late hour.

Quick checklist

Done	Task
[ ]	Midterm paper attempted and marked
[ ]	Cheat sheet: classes, constructors, static
[ ]	Cheat sheet: inheritance, super, constructor order
[ ]	Cheat sheet: overriding vs overloading, polymorphism
[ ]	Traced Overridden_Methods 1/2/3 and Polymorphism_Rectangle
[ ]	T1 paper attempted and marked
[ ]	Wrote Student/Account and Shape programs by hand
[ ]	Reviewed R_ papers on weak topics
[ ]	Final cheat sheet read twice